50 YEARS LATER: SERVING A SPACE AGENCY CLIENT – THE LAWYER'S ROLE IN INTERNATIONAL SPACE COOPERATION

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Last fall, I participated in the 2nd Eilene M. Galloway Symposium on Critical Issues in Space Law (December 6, 2007) by speaking on the topic of "Serving the Space Agency and other Space Clients: Achieving Effective International Cooperation Through Legal Advice." Several participants found my presentation interesting and thought the subject would be of interest to a broader audience. Thus, I have elaborated on the topics of that presentation and put them in the broad context of international law practice at the National Aeronautics and Space Administration (NASA) in this Article.

Lawyers at NASA provide legal advice and services on a broad variety of subjects, ranging from contracts and procurement to personnel, ethics, intellectual property, and commercial law. The lawyers in the International Law Practice Group (ILPG) are responsible for providing legal advice in direct support of NASA's international programs. The legal advice covers, among other issues, drafting and negotiating international agreements regarding international space operations, exploration, science and aeronautics cooperation, legal issues that arise in implementation of these agreements, space law, domestic laws that implicate foreign involvement in NASA activities, counter-intelligence, and export control.

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The ILPG — in conjunction with other Headquarters legal practice groups and chief counsel offices at NASA field centers — supports the General Counsel in providing the Administrator and NASA programs, the Office of External Relations, Mission Directorates, and Centers with advice and counsel on international legal issues affecting the Agency. We advise the different components of NASA as to what the relevant law provides and help them to resolve all competing interests before proceeding with international programs. In effect, we provide tactical and strategic advice on how NASA moves forward in these areas.

To illustrate this process, this article discusses the provision of legal advice and counsel in two specific program areas of great importance to NASA's mission, each of which is at very different stages of development. First, the article focuses on space cooperation with Russia in the mature context of the *International Space Station* (*ISS*) program. Second, it examines briefly the evolving multilateral dimensions of the U.S. Space Exploration Policy.

I. COOPERATION WITH RUSSIA ON THE INTERNATIONAL SPACE STATION

Over the past few years, NASA and the International Partners have been concerned with final assembly and use of the ISS. The situation is legally mature. The United States, Japan, Canada, Russia, and Member States of the European Space Agency entered into the Agreement Among the Government of Canada, Governments of Member States of the European Space Agency, the Government of the Russian Federation and the Government of the United States of America Concerning Cooperation on the Civil International Space Station, signed at Washington, January 29, 1998 (Intergovernmental Agreement or IGA)[1] that provides the framework for cooperation. Moreover, the United States entered into Memoranda of Understand-

[1] Agreement Among the Government of Canada, Governments of the Member States of the European Space Agency, the Government of Japan, the Government of the Russian Federation and the Government of the United States of America Concerning Cooperation on the Civil International Space Station, Jan. 29, 1998 [hereinafter IGA], *available at* ftp://ftp.hq.nasa.gov/pub/pao/reports/1998/IGA.html.

ing (MOU) concerning cooperation on the International Space Station (ISS) with the Canadian Space Agency (signed January 29, 1998), the European Space Agency (signed January 29, 1998), the Russian Federal Space Agency (Roscosmos or RSA) (signed January 29, 1998) and the Government of Japan (signed February 24, 1998).[2] Attorneys from the predecessor office to the ILPG were members of the United States Government delegation that negotiated the IGA and the NASA team that negotiated the MOUs. These agreements establish an array of governing boards that address a wide variety of assembly and operational issues and make various decisions on cooperation. The ILPG — often in consultation with the Department of State's Office of the Legal Adviser and/or the Johnson Space Center's (JSC) Office of Chief Counsel and, at times, in coordination with the International Partners' counsel — provides legal advice about the IGA, MOUs, Implementing Arrangements, and occasionally addresses legal questions that arise in the work of the governing boards.

This Section addresses, in particular, the legal framework, below the level of the NASA-Russian Federal Space Agency Memorandum of Understanding, that governs the relationship between the two space agencies and at the manner in which a U.S. nonproliferation statute has affected this relationship.

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[2] Memorandum of Understanding Between the National Aeronautics and Space Administration of the United States of America and the Canadian Space Agency Concerning Cooperation on the Civil International Space Station, Jan. 29, 1998, *available at* ftp://ftp.hq.NASA.gov/pub/pao/reports/1998/nasa_csa.html. Memorandum of Understanding Between the National Aeronautics and Space Administration of the United States of America and the European Space Agency Concerning Cooperation on the Civil International Space Station, Jan. 29, 1998, *available at* ftp://ftp.hq.nasa.gov/pub/pao/reports/1998/nasa_esa.html. Memorandum of Understanding Between the National Aeronautics and Space Administration of the United States of America and the Russian Space Agency Concerning Cooperation on the Civil International Space Station, Jan. 29, 1998 [hereinafter NASA-Roscosmos MOU] *available at* ftp://ftp.hq.nasa.gov/pub/pao/reports/1998/nasa_russian.html. Memorandum of Understanding Between the National Aeronautics and Space Administration of the United States of America and the Government of Japan Concerning Cooperation on the Civil International Space Station, Feb. 24, 1998, *available at* ftp://ftp.hq.nasa.gov/pub/pao/reports/1998/nasa_japan.html.

A. 1990s – 2005

The Protocol Including Terms, Conditions and Assumptions, Summary Balance of Contributions and Obligations to International Space Station (ISS) and Resulting Rights of NASA and RSA to ISS Utilization Accommodations and Resources, and Flight Opportunities[3] (Balance Agreement) was concluded in 1996. While the Balance Agreement predates the IGA and the NASA-Roscosmos MOU, by its terms it became an implementing arrangement[4] under those agreements. Its basic premise — and this understanding carries over to present day relations between Russia and the other ISS Partners — is that the ISS is made up of two segments, the Russian Segment and the U.S. on-orbit segment (USOS).[5] Each side retains the use and benefits of the elements it provides (*e.g.*, launch of elements, share of crew, crew supplies, life support consumables, spares and payloads) and is responsible for support of its own elements ("keep what they bring"), unless otherwise provided for in the Balance Agreement.[6] The Balance Agreement addresses provisions of goods and services on a comparable basis ("balanced" basis) that cross the interface between the Russian Segment and the USOS. In the Balance Agreement, NASA and Roscosmos agree to mutual provision of goods and services covering the life of the ISS Program, without exchange of funds. The Balance Agreement (recognizing that these issues would also be addressed in the NASA-Roscosmos MOU), addresses such issues as flight opportunities and crew time, crew rotation and rescue, propellant delivery, the status of a Russian-built U.S. component of the *ISS*, the Functional Cargo Block (FGB, also known as

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[3] The Protocol Including Terms, Conditions and Assumptions, Summary Balance of Contributions and Obligations to International Space Station (ISS) and Resulting Rights of NASA and RSA to ISS Utilization Accommodations and Resources, and Flight Opportunities (June 11, 1996) [hereinafter Balance Agreement] (on file with author).

[4] The IGA provides that Cooperating Agencies (here, NASA and Roscosmos) may enter into arrangements to implement the MOUs – "implementing arrangements." *See* 1988 IGA, *supra* note 1, at art. 4. The MOUs are subject to the IGA and the implementing arrangements are consistent with and subject to the MOUs.

[5] Balance Agreement, *supra* note 3, at para. 1.

[6] *Id.* at para. 4.

Zarya), and NASA's launch of the originally envisioned Russian Science Power Platform.[7]

NASA also has a contract with Roscosmos dating back to *Shuttle-Mir* cooperation (since 1993). The terms of the contract provide, *inter alia*, for NASA to pay Roscosmos for the provision of certain goods and services that NASA is responsible for providing to the Partnership under the IGA and MOUs.[8]

Since the original Balance Agreement and NASA-Roscosmos contract were concluded, many things have changed. First, Russia's severe financial difficulties in the 1990s, coupled with engineering challenges on the U.S. side, resulted in significant delays in the early assembly of the *ISS*. Second, the Iran Nonproliferation Act (INA) was enacted in 2000. The INA prohibited payments, in cash or in kind, related to the *International Space Station* by the United States Government (and its contractors) to Roscosmos and many other Russian entities, except for payments required to be made under the terms of contracts and agreements in effect on January 1, 1999.[9] Third, the

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[7] *Id.* at paras. 11, 13, 14, 17, 19, & 20.

[8] Contract NAS15-10110.

[9] Iran Nonproliferation Act of 2000, Pub. L. No. 106-178, 114 Stat. 38, 50 U.S.C. § 1701 (Jan. 4, 2000), at note [hereinafter INA].

INA provides in relevant part:

> SEC. 6. RESTRICTION ON EXTRAORDINARY PAYMENTS IN CONNECTION WITH THE INTERNATIONAL SPACE STATION.
>
> (a) Restriction on Extraordinary Payments in Connection With the International Space Station.—Notwithstanding any other provision of law, no agency of the United States Government may make extraordinary payments in connection with the International Space Station to the Russian Aviation and Space Agency, any organization or entity under the jurisdiction or control of the Russian Aviation and Space Agency, or any other organization, entity, or element of the Government of the Russian Federation, unless, during the fiscal year in which the extraordinary payments in connection with the International Space Station are to be made, the President has made the [nonproliferation] determination described in subsection (b), and reported such determination to the [appropriate Congressional Committees]....

Id. at § 6.

> SEC. 7. DEFINITIONS.
>
> For purposes of this Act, the following terms have the following meanings:
>
> (1) Extraordinary payments in connection with the international space station.—The term 'extraordinary payments in connection with the Inter-

grounding of the *Space Shuttle* fleet following the *Space Shuttle Columbia* tragedy further delayed completion of the *ISS* Assembly. Fourth, both sides changed their planned contributions, thus affecting the end-state configuration of the *ISS*.

The Addendum to the 1996 Balance of Contributions Protocol for Services to be Provided through December 2005 and Crew Rescue through April 2006 (First Addendum)[10] was negotiated in 2004 and entered into force in January 2005. It was intended to be an interim step in an update process, beginning to account for the changes just mentioned. From a program perspective, NASA and Roscosmos needed to adjust the Balance Agreement to address immediate operational needs — crew rotation responsibilities through 2005, crew rescue through early 2006, and Russian launch and return of NASA cargo, also a critical need because of the then-grounded *Shuttle*.

The First Addendum was negotiated when the INA was in force with no relief. NASA lawyers helped to ensure that the Addendum did not result in payments "in cash or in kind" to Roscosmos that were prohibited by the INA. Additionally, NASA lawyers demonstrated to lawyers for the State Department and representatives of other Executive branch agencies that the First Addendum did not result in such payments, as State Department and interagency approval of the agreement through the "Circular 175 process"[11] was necessary for the First Addendum to enter into force.

NASA lawyers explained that much of the First Addendum reiterated and confirmed existing obligations under the Balance

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> national Space Station' means payments in cash or in kind made or to be made by the United States Government—
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> (A) for work on the International Space Station which the Russian Government pledged at any time to provide at its expense; or
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> (B) for work on the International Space Station, or for the purchase of goods or services relating to human space flight, that are not required to be made under the terms of a contract or other agreement that was in effect on January 1, 1999, as those terms were in effect on such date....

Id. at § 7.

[10] Balance Agreement , *supra* note 3, at 2005 Addendum [hereinafter First Addendum].

[11] 22 C.F.R. Part 181. 11 U.S. DEPARTMENT OF STATE FOREIGN AFFAIRS MANUAL 720 (Sep. 25, 2006), *available at* http://www.state.gov/documents/organization/88317.pdf

Agreement, and that these continued beyond the originally-planned Assembly Complete date. For example, NASA and Roscosmos agreed that Roscosmos would meet its Balance Agreement *Soyuz* crew rotation obligations sequentially, rather than out of order, which ensured American presence on the *ISS* before *Shuttle* return to flight. In addition, the First Addendum provided for a balanced trade. Under a 1998 amendment to the 1993 NASA-Roscosmos contract, Roscosmos agreed to provide NASA with 4000 research crew hours for close to $60 million. By late 2004, it was clear that NASA was not going to be able to use the crew hours as originally planned. Because of the grounding of the *Shuttle*, NASA needed upmass — that is, for Roscosmos to launch NASA cargo. So the Addendum provided that NASA would relinquish the remaining contract crew hours and that, until the Shuttle returned to flight, Roscosmos would launch up to 1.7MT of NASA cargo (such as crew provisions and critical spares) and would return a small amount of downmass. In January 2005, Circular 175 authority to conclude the First Addendum was obtained and the agreement entered into force.

B. 2005 – 2007

The First Addendum was only an interim measure. Accordingly, further amendments to the Balance Agreement, as well as to the NASA-Roscosmos contract mentioned above, to cover, *inter alia*, purchase of *Soyuz* crew transportation and rescue services and Progress cargo services, were necessary. In addition to the changes outlined above, in 2005, the United States government decided to reduce the number of *Shuttle* flights for the *ISS* to 18 or fewer and obtained consensus from the International Partners to do so.

Moreover, obtaining relief from the restrictions of the Iran Nonproliferation Act was necessary before the Balance Agreement and the NASA-Roscosmos contract could be further amended. Throughout 2005, NASA's lawyers worked with a NASA team and an interagency group, chaired by National Security Council staff, to develop, propose to Congress and then work to enact an amendment to the Iran Nonproliferation Act that would adjust the INA's prohibition on "extraordinary pay-

ments" by the United States Government to Roscosmos and other proscribed Russian entities. In November 2005, the Iran Nonproliferation Amendments Act amended the INA to, *inter alia*, allow for payments to Roscosmos and other proscribed Russian entities until January 1, 2012, for work and services to be performed before that date related to the *ISS*.[12]

In December 2005, using the new Iran and Syria Nonproliferation Act (ISNA)[13] exception, NASA concluded a short-term modification to the NASA-Roscosmos contract discussed above. JSC attorneys were closely involved in drafting and negotiating this amendment. This amendment covered immediate needs of the ISS program, including crew transportation and rescue for the next U.S. crewmember on a *Soyuz* and delivery of U.S. cargo to the ISS on the next two Progress flights.

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[12] Iran Nonproliferation Amendments Act of 2005, Pub. L. No. 109-112, 119 Stat. 2366 (Nov. 22, 2005), amended the INA to, *inter alia*, adjust the definition of extraordinary payments as follows:

> Sec.7. Definitions.
>
> For purposes of this Act, the following terms have the following meanings:
>
> (1) Extraordinary payments in connection with the international space station. The term 'extraordinary payments in connection with the International Space Station' means payments in cash or in kind made or to be made by the United States Government-- ...
>
> (B) for work on the International Space Station, or for the purchase of goods or services relating to human space flight, that are not required to be made under the terms of a contract or other agreement that was in effect on January 1, 1999, as those terms were in effect on such date, *except that such term does not mean payments in cash or in kind made or to be made by the United States Government prior to January 1, 2012, for work to be performed or services to be rendered prior to that date necessary to meet United States obligations under the Agreement Concerning Cooperation on the Civil International Space Station, with annex, signed at Washington January 29, 1998, and entered into force March 27, 2001, or any protocol, agreement, memorandum of understanding, or contract related thereto.* Due to other changes in the statute, the statute was renamed the Iran and Syria Nonproliferation Act.

Id. at § 7 (emphasis added). INA, *supra* note 9, § 7.

[13] Due to other changes in the statute, the Iran Nonproliferation Amendments Act of 2005 was renamed the Iran and Syria Nonproliferation Act [hereinafter ISNA]. *See* Iran Nonproliferation Amendments Act of 2005, *supra* note 12, § 4(e)(1).

Negotiations of the Second Addendum to the Implementing Arrangement Entitled "Protocol Including Terms, Conditions and Assumptions, Summary Balance of Contribution and Obligations to International Space Station (ISS) and Resulting Rights of NASA and RSA to ISS Utilization Accommodations and Resources, and Flight Opportunities" (Second Addendum)[14] took place in early 2006. NASA lawyers helped to draft NASA's proposed agreement. ILPG also helped the Office of External Relations to seek and obtain "Circular 175" authority to negotiate and conclude the agreement. The Second Addendum was intended to partially rebalance NASA and Roscosmos' obligations over a longer period of time to take into account the factors described above. Critical for NASA was eliminating its obligation to launch the Russian Science Power Platform, given the decreased number of Shuttle flights. Also key was reducing NASA's obligation to provide metric tons of goods to the *ISS*, given the limited number of *Shuttle* flights remaining and the significant expense involved in purchasing substantial space aboard U.S. commercial providers' vehicles and International Partners' vehicles, as they become available. Critical for Roscosmos was ensuring that it has an adequate supply of electrical power to assemble, maintain, and operate the Russian Segment for the life of the Station, given that the Russian Science Power Platform could not have been flown before the planned retirement of the *Shuttle*. From a program perspective, both sides achieved their objectives.

One role of ILPG was to help negotiate appropriate agreement text and provide technical drafting assistance in amending the Balance Agreement. NASA lawyers also worked to ensure that the Addendum was consistent with the ISNA. For most elements of the barter, consistency with the legislation was a matter of providing that payment for, and provision of, the goods and services would all take place by December 31, 2011. Such elements included, for example, Roscosmos' agreement to provide crew rotation and rescue services and logistics support by contract, and NASA's agreement to provide stowage and com-

[14] Balance Agreement, *supra* note 3, at 2006 Addendum [hereinafter Second Addendum].

munications services. On two elements of the barter — pertaining to NASA providing Roscosmos with electrical power and also with limited habitation services — continuing the activities beyond 2011 was necessary to conclude the deal. NASA's lawyers helped to establish that NASA could rely on the provision remaining from the original Iran Nonproliferation Act under which NASA is permitted to continue to make payments for work on the *ISS* required to be made under the terms of a contract or agreement in effect on January 1, 1999.[15] NASA provision of both habitation and electrical power is provided for in the 1998 NASA-Roscosmos MOU,[16] an agreement in effect on January 1, 1999. As a continuation of MOU requirements, provision of these services through 2015 was thus permissible.

Ad referendum agreement was reached on the Second Addendum in time for the March 2006 *ISS* Heads of Agency meeting so that the new Assembly schedule, which did not include launch of the Russian Science Power Platform, could be announced at that meeting. The Second Addendum was signed and entered into force in July 2006.

The approach of limiting NASA's provision of goods and services to those that would be provided by Roscosmos prior to Janaury 1, 2012, and that would be paid for prior to that date, was later followed in the amendments to the NASA-Roscosmos contract for crew rotation and rescue services, cargo services, and other Russian-unique goods and services. The principal contract extension, providing for purchase of crew rotation (launch and return) and rescue services until the fall of 2011, cargo services through 2011 and miscellaneous other Russian-unique goods and services was completed in the spring of 2007. It is worth approximately $719 million.[17] ILPG worked with Headquarters' Contract and Procurement Law Group and JSC lawyers on this modification to address contract law and regulations issues, as well as ISNA issues.

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[15] *See* INA, *supra* note 9, at sec. 7(1)(B) (language quoted in footnote 9; this language remained in place in the ISNA).

[16] NASA-Roscosmos MOU, *supra* note 2, at arts. 3.2 & 6.1.b.20.

[17] Michael Braukus, *NASA Extends Contract With Russia's Federal Space Agency* (Apr. 9, 2007), *available at* www.nasa.gov/home/hqnews/2007/apr/HQ_C07-18_Roscosmos.txt.

C. 2007 - present

The ISS Partners have agreed that crew size on the ISS should expand from three crew members to six crew members in Spring 2009. The United States is responsible for the crew transportation and rescue services for three USOS crew members on the *ISS* (whether they be U.S., Canadian, Japanese, or European crew members) beginning at this time. These responsibilities are built into the NASA-Roscosmos contract through the Spring 2011–Fall 2011 flight increment, given the limitations on payments in what is now the Iran, North Korea, and Syria Nonproliferation Act (INKSNA).[18] In technical discussions with Roscosmos in late 2007, however, the Russians made clear that it took approximately 36 months to build a *Soyuz* capsule and that NASA and Roscosmos would need to have a contract in place covering the Fall 2011 – Spring 2012 increment no later than the Fall of 2008 if NASA expected to rely on a *Soyuz* vehicle for that flight.

In the Winter and early Spring of 2008, there were extensive discussions within the Administration, led by officials from the Office of Science and Technology Policy, about the possibility of seeking legislative relief of INKSNA's terms and what an Administration proposal would contain. ILPG lawyers drafted various versions of a possible amendment and its related analysis, and participated in the various discussions. In April 2008, NASA Administrator Mike Griffin sent a narrowly-crafted proposal to Congress. In addition to maintaining the key nonproliferation provisions of INKSNA, the proposal would extend INKSNA's exception to the prohibition on "extraordinary payments" to the Russian government and Russian entities for goods or services relating to the *ISS* from January 1, 2012 to the end of the life of the *ISS*. The proposal would exclude from the exception any payments after December 31, 2011, for cargo services provided by a *Progress* vehicle. The proposal would also exclude from the exception payments for crew transportation or

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[18] The North Korea Nonproliferation Act of 2006, P.L. 109-353, 120 Stat. 2015 (Oct. 13, 2006), amended ISNA to include North Korea and renamed the statute accordingly. No substantive changes were made to sections 6 and 7.

rescue services provided by a Soyuz vehicle once (1) the U.S. *Orion Crew Exploration Vehicle* reaches Full Operational Capability or (2) a U.S. commercial provider of crew transportation and rescue services demonstrates the capability to meet ISS mission requirements.[19]

NASA officials have had detailed discussions with the staff of Committees interested in the legislation, including the Senate Foreign Relations Committee, House Foreign Relations Committee, House Subcommittee on Space and Aeronautics, Senate Commerce Committee and both Chambers' appropriations committees. Several legislative possibilities have been proposed. ILPG has been involved in these discussions and in analyzing the legislative possibilities. On June 9, 2008, Senators Joe Biden and Richard Lugar introduced the Administration's proposal in the Senate.[20] Ultimately, after the personal intervention of the Administrator, the Deputy Administrator, and other senior Administration officials with key Congressmen and Senators, a provision was included in the Consolidated Security, Disaster Assistance and Continuing Appropriations Act, 2009 (the Continuing Resolution) that simply changed the date of the INKSNA exception added by the 2005 Iran Nonproliferation Amendments Act[21] from "January 1, 2012" to "July 1, 2016".

The Continuing Resolution was signed into law by President Bush on September 30, 2008.[22] NASA and Roscosmos will now work together on an appropriate amendment or amendments to the NASA-Roscosmos contract[23] that provides for further crew transportation and rescue services and other Russia-unique services (*e.g.*, sustaining engineering and spares).

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[19] *See* International Space Station Payments Act of 2008, S. 3103, 110th Cong. (2008), as introduced by Senators Biden and Lugar at the Administration's request; *see also* Letter from Michael Griffin, NASA Administrator, to Hon. Joseph R. Biden, Chairman, Committee on Foreign Relations, U.S. Senate, Washington D.C., (April 11, 2008), and enclosed Sectional Analysis. CONG. REC. SENATE, p.S5392 (June 9, 2008) (identical letters were sent to the relevant Congressional Committees and Subcommittees' Chairmen and Ranking Members in April 2008).

[20] *Id.* (Congressional Record discussion).

[21] *See* Iran Nonproliferation Amendments Act of 2005, *supra* note 12.

[22] Consolidated Security, Disaster Assistance and Continuing Appropriations Act, 2009, Pub. L. No. 100-329 (September 30, 2008) [hereinafter Continuing Resolution].

[23] *See* notes 8 & 17.

II. U.S. SPACE EXPLORATION POLICY – MULTILATERAL COOPERATION

The U.S. Space Exploration Policy, announced by President Bush in January 2004,[24] states that its fundamental goal is to "advance U.S. scientific, security and economic interests through a robust space exploration program," and specifically calls on the Administrator to "pursue opportunities for international participation to support U.S. space exploration goals."[25] At this stage of planning to return to the Moon, the focus of NASA's attention on the U.S. Space Exploration program is developing the transportation systems – the crew exploration vehicle, the launch vehicles, and the lunar lander – to return to the Moon. NASA, however, also has expended substantial efforts and attention on fostering international cooperation. As the Administrator has stated repeatedly, NASA

> will not attempt to prescribe the manner of participation of any of our potential partners. We will work with others to define an exploration architecture suitable to all, and we will identify those portions of the task that we are willing and able

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[24] The White House, *A Renewed Spirit of Discovery* [hereinafter U.S. Space Exploration Policy], www.whitehouse.gov/space/renewed_spirit.html (last visited Oct. 23, 2008). This policy was confirmed by the U.S. Congress, with strong bipartisan support, with the passage of the NASA Authorization Act of 2005, Pub. L. No. 109-155, 119 Stat. 2946 (Dec. 30, 2005).

[25] The U.S. Space Exploration Policy states that in support of this goal,

> the United States will:
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> • Implement a sustained and affordable human and robotic program to explore the solar system and beyond;
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> • Extend human presence across the solar system, starting with a human return to the Moon by the year 2020, in preparation for human exploration of Mars and other destinations;
>
> • Develop the innovative technologies, knowledge, and infrastructures both to explore and to support decisions about the destinations for human exploration; and
>
> • Promote international and commercial participation in exploration to further U.S. scientific, security, and economic interests.

Id.

> to accomplish with the funding we can provide. We expect that others will do the same.[26]

An integral part of U.S. space exploration will be the exploration of the Moon and other destinations in cooperation with international partners.[27] NASA's cooperation with international partners has included meeting United States Government commitments to assemble the ISS and various bilateral initiatives with respect to both science and exploration objectives.

A cornerstone of NASA's international cooperation has been a multilateral initiative. Several years ago, NASA initiated a dialogue with representatives of 13 other international space agencies, a group which informally became known as the "Global Exploration Strategy."[28] In May 2007, the 14 space agencies agreed on the text and public release of "The Global Exploration Strategy: The Framework for Coordination,"[29] informally known as the "Framework Document." The Framework Document presents a vision for robotic and human space exploration, focusing on destinations within the solar system where we may one day live and work.[30] The Framework document explains why a sustainable but affordable agenda of globally coor-

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[26] Michael T. Griffin, NASA Administrator, Address Before the Parliamentary Group on Space at the French National Assembly (June 5, 2008), www.nasa.gov/news/speeches/admin/index.html.

[27] U.S. Space Exploration Policy, *supra* note 24.

[28] *The Global Exploration Strategy: The Framework for Coordination* [hereinafter *Framework Document*] (May 2007), *available at* www.nasa.gov/pdf/178109main_ges_framework.pdf 1 [hereinafter *Framework Document*]. For a detailed discussion about the development of the international group and "The Global Exploration Strategy: The Framework for Coordination", *see* Gibbs *et. al.*, *The Global Exploration Strategy: Developing a Framework for International Coordination and Cooperation*, IAC-07-B3.1.08 (Sept. 2007).

In this context, "space agencies" refer to government organizations responsible for space activities. Participants came from the following 14 space agencies: Australian Commonwealth Scientific and Industrial Research Organization (CSIRO), British National Space Centre (BNSC), Canadian Space Agency (CSA), China National Space Administration (CNSA), French National Center for Space Studies (CNES), German Aerospace Center (DLR), Indian Space Research Organization (ISRO), Italian Space Agency (ASI), Japan Aerospace Exploration Agency (JAXA), Korea Aerospace Research Institute (KARI), National Aeronautics and Space Administration (NASA), National Space Agency of Ukraine (NSAU), Russian Federal Space Agency (Roscosmos), and European Space Agency (ESA). *Framework Document, supra* note 28.

[29] *Framework Document, supra* note 28.

[30] *Id.* at 2.

dinated space exploration can serve society through exploring five themes:

- New Knowledge in Science and Technology;
- A Sustained Presence – Extending Human Frontiers;
- Economic Expansion;
- A Global Partnership; and
- Inspiration and Education.[31]

The Framework Document also makes the case for a formal, though non-binding and voluntary, mechanism that could "help coordinate global space exploration by: providing a forum for participants to discuss their interests, objectives and plans in space exploration; and promoting interest and engagement in space exploration activities throughout society."[32] In November 2007, agreement among many of the 14 space agencies was reached on the International Space Exploration Coordination Group Terms of Reference (ISECG ToR).[33]

The ILPG provides legal support to the NASA team that participates in this ongoing multilateral process. This Section will thus address several legal issues raised by the work on the Framework Document and the ISECG Terms of Reference.

The Framework Document is a policy statement, not a legally binding instrument. As a consensus-based document, various aspects of the Framework Document raise possibilities but leave open legal issues.[34] ILPG worked to make sure the NASA

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[31] *Id.* at ch. 2.

[32] *Id.* at ch. 6.

[33] International Space Exploration Coordination Group Terms of Reference (Nov. 2007) [hereinafter ISECG ToR], *available at* http://esamultimedia.esa.int /docs/exploration /InternationalCoordination/ISECG_ToR.pdf.

[34] Underpinning the Framework Document and subsequent cooperation are the four core outer space treaties that establish a legal framework for cooperation related to outer space, including the Moon and other celestial bodies: Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and other Celestial Bodies, Jan. 27, 1967, T.I.A.S. 6347, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty]; Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space, Apr. 22, 1968, U.N. GAOR, 22nd Sess., Supp. No. 16, U.N. Doc. A/6716 (1968), 19 U.S.T. 7570, 1968 U.S.T. LEXIS

team did not get locked into policy positions which would require particular legal interpretations of existing international agreements or particular next steps. Examples of these areas of concern follow.

The Framework Document embraces the theme of economic expansion and talks about "further commercial expansion into space. As space exploration extends to the Moon and Mars, there will be potential opportunities for companies to provide crew and cargo transportation services, telecommunications and navigation systems, and space-based resource extraction and processing capabilities."[35] The Framework Document also mentions real and virtual space tourism on the Moon or Mars. There will likely be clear international consensus that some of these uses are authorized under the 1967 Outer Space Treaty. Some of the uses — depending on how they are carried out — may raise legal issues. Moreover, the section on economic expansion states that for business to be confident about investing, it needs[36] a "common understanding on such difficult issues as property rights and technology transfer."[37]

Whether there will be an international consensus on what property rights exist or should exist with respect to the Moon and other celestial bodies is an open question and there are widely different views as to what the Outer Space Treaty allows (*e.g.*, exclusive use of real property, ability to use minerals and other substances found in or on real property). Art. II of the Outer Space Treaty provides that "[o]uter space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means."[38] Art. VI of the Outer Space Treaty provides, in relevant part, that

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584; Convention on International Liability for Damage Caused by Space Objects, Mar. 29, 1972, 961 UNTS 187; Convention on the Registration of Objects Launched into Outer Space, Jan., 14, 1975, U.N. GAOR, 29th Sess., Supp. No. 31, U.N. Doc. A/9631 (1975), 28 U.S.T. 695, 1975 U.S.T. LEXIS 552.

[35] *Framework Document, supra* note 28, at 11.

[36] "For business to be confident about investing, it needs the certainty of a long-term commitment to space exploration, the opportunity to introduce its ideas into government thinking, and the rule of law." *Id.*

[37] *Id.*

[38] Outer Space Treaty, *supra* note 34, at art. II.

> States Parties to the Treaty shall bear international responsibility for national activities in outer space, including the Moon and other celestial bodies, whether such activities are carried on by governmental agencies or by non-governmental entities, and for assuring that national activities are carried out in conformity with the provisions set forth in the present Treaty. The activities of non-governmental entities in outer space, including the Moon and other celestial bodies, shall require authorization and continuing supervision by the appropriate State Party to the Treaty.[39]

Some states are likely to take the view that Art. II, in conjunction with Art. VI of the Outer Space Treaty, prohibits private ownership of real property in outer space. In other words, they would read together the prohibition on national appropriation in Art. II and the Art. VI requirement that all activities of non-governmental entities shall be conducted under the auspices of state authority. They would thus conclude that any private appropriation — "ownership" of real property in outer space — requires state action and thus is a prohibited national appropriation.

Other states are likely to take the view that Art. II does not prohibit private ownership. They would rely on the express language of Art. II and argue that, if the negotiators had intended to prohibit non-governmental entities from appropriating property, they would have said so explicitly.

As a second example, change to the International Traffic in Arms Regulations[40] to make transfer of space-related technology and goods easier was a theme of some of the Global Exploration Strategy space agencies. The Framework document simply speaks of the need for a common understanding regarding "technology transfer."[41]

As a third example, environmental protection in outer space was discussed. The chapter on the Moon ended up with a statement that "the lunar environment is both fragile and spe-

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[39] *Id.* at art. VI.
[40] 22 C.F.R. Parts 120 through 130.
[41] *See* text associated with notes 36 & 37.

cial; we must protect and preserve it even as we explore it"[42] and the chapter about Mars contains a parallel statement[43]. Moreover, the Framework Document points out that "[c]omplex issues such as the protection of areas of scientific importance may arise and can be discussed before they block progress."[44] The Outer Space Treaty already provides some protections. Art. IX of the Outer Space Treaty requires States Parties to

> pursue studies of outer space, including the Moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, ...[to] adopt appropriate measures for this purpose.[45]

For each of the three examples, the language used in the Framework Document is sufficiently neutral and noncommittal such that NASA was able to join consensus without a new or updated United States government policy. The Framework Document leaves open what appropriate next steps might be but proposes that the multilateral coordination mechanism will provide a forum to discuss these issues.[46] The Framework Document also leaves open where NASA will address these issues and other international issues arising as space exploration develops – *e.g.* through national laws, the ISECG and other multilateral fora, and/or new or amended multilateral treaties. The ISECG ToR provide for a broad and strategic scope of ISECG activities, certainly including these issues if Participating Agencies so decide.[47]

The Framework Document and the work of the ISECG set the stage for bilateral and perhaps multilateral cooperation through legally binding agreements, among other objectives.

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[42] *Framework Document, supra* note 28, at 19.

[43] *Id.* at 21.

[44] *Id.* at 13.

[45] Outer Space Treaty, *supra* note 34.

[46] *Framework Document, supra* note 28, at 11, ch. 6.

[47] ISECG ToR, *supra* note 33, at I.2. ISECG ToR I.2, in particular, provides that one area for initial consideration is the "assessment of the requirement for any relevant international legal agreements." *Id.*

For NASA, this will mean agreements concluded under NASA's Space Act authority[48] with United States government interagency coordination led by the State Department. Neither the Framework Document nor the establishment of the ISECG set the stage for one over-arching multilateral agreement, as with the *International Space Station* model. Rather, they foresee a more flexible and open approach. As the ISECG further matures and NASA continues bilateral and multilateral cooperation, there will be many issues and agreements about which the ILPG will engage with NASA's Exploration Systems Mission Directorate, Science Mission Directorate, Space Operations Mission Directorate, Office of External Relations, other NASA policy and program offices, other U.S. agencies, and NASA's international counterparts.

CONCLUSION

As the foregoing discussion indicates, the role of the international lawyers at NASA is broad and varied. NASA lawyers are an integral part of NASA's efforts to structure cooperation in international programs. This article is not meant to be an exhaustive review but to provide insight into two examples of NASA lawyers' contributions to NASA's international cooperative programs.

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[48] National Aeronautics and Space Act of 1958, as amended, 42 U.S.C. § 2451 *et seq.*

THE EXTRA-TERRITORIAL REACH OF U.S. PATENT LAW ON SPACE-RELATED ACTIVITIES: DOES THE "INTERNATIONAL SHOE" FIT AS WE REACH FOR THE STARS?

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I. INTRODUCTION

The law of a nation has its jurisdictional roots planted in such nation's territories and possessions. Thus, the territorial jurisdiction of a national court over a person or a case or controversy concerning an activity typically has its limits defined by the geographical boundaries of the territory to which a particular person resides or an activity originates. Historically, national patent laws have been no exception to this general rule.

Extra-territorial jurisdiction is a limited expansion of the territorial authority of a nation and its courts. A national court may exercise extra-territorial jurisdiction in other countries via a treaty or by such nation's own ministers or consuls in foreign lands. Extra-territorial jurisdiction (or "reach") of a national court should be based at least in part on a nexus between the

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